MID-TERM EXAMINATION

(Programme: B.Tech (CSE/ CSE-AI/ IT/ AI&ML/ ECE/ ECE-AI/ MAE/DMAM) (Semester: II)

(March 2026) OFF LINE mode

Subject Code: BAI 103	**Subject: Introduction To Data Science**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**
	a) Define data science. Write its applications any four.	
	b) Explain indexing, advance indexing and slicing in NumPy arrays.	
	c) Describe the steps involved in data wrangling with its definition.	
	d) Write a python program to create NumPy array for 1D, 2D and 3D array.	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) What is data exploration? Explain its steps and importance.	
	b) Explain the life cycle of data science in detail.	
	c) Explain the process of data filtering and data merging.	
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	a) Write a python program to perform numerical operations like (add, subtract, multiply and divide) on NumPy arrays.	
	b) What are ethics in data science? Explain the 5 Cs of data science.	
	c) Define data cleaning. Explain the steps involved in data cleaning process along with advantages and disadvantages.	